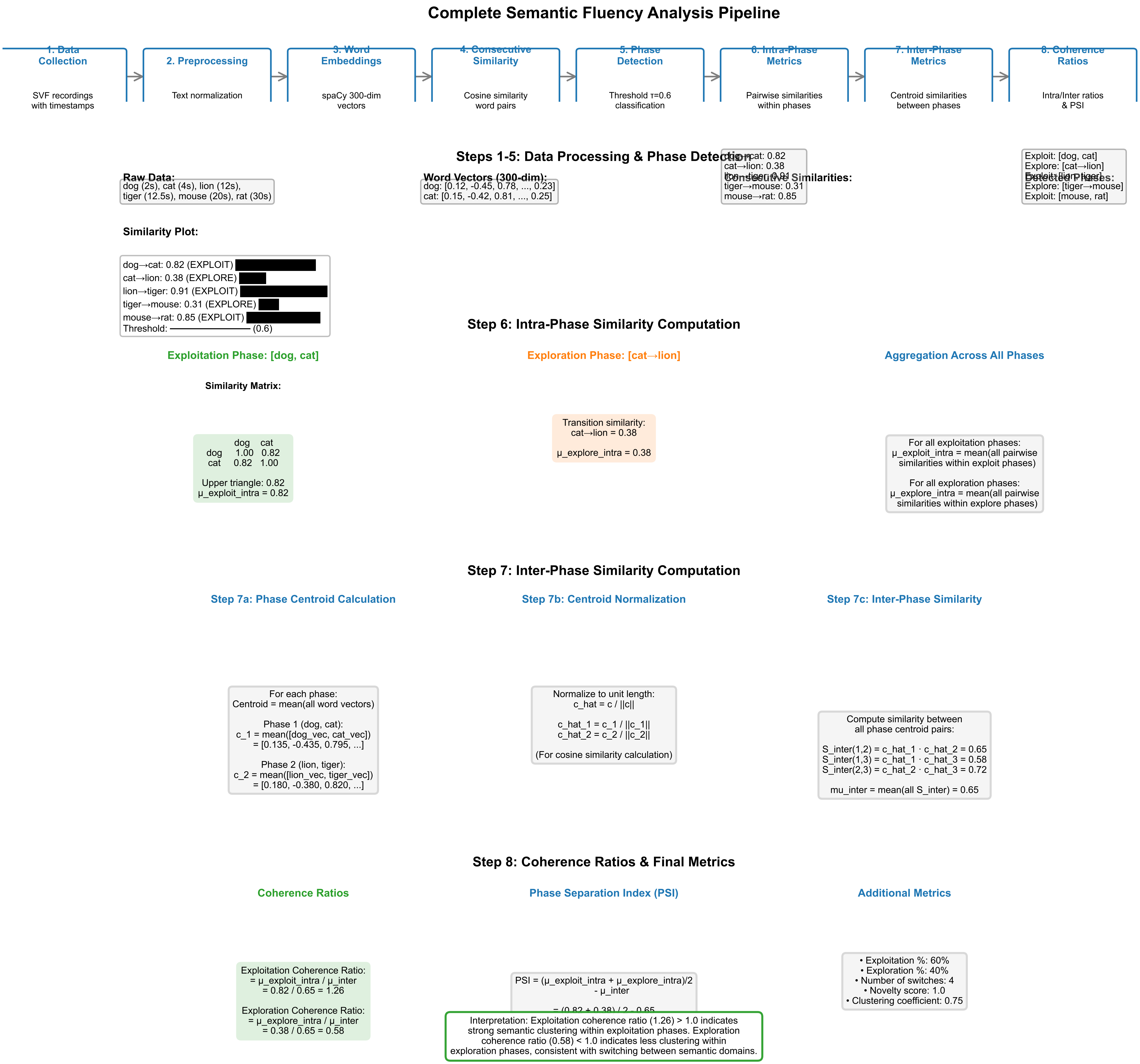
Complete Semantic Fluency Analysis Pipeline
1. Data Collection
SVF recordings with timestamps
2. Preprocessing
Text normalization
3. Word Embeddings
spaCy 300-dim vectors
4. Consecutive Similarity
Cosine similarity word pairs
5. Phase Detection
Threshold τ=0.6 classification
6. Intra-Phase Metrics
Pairwise similarities within phases
7. Inter-Phase Metrics
Centroid similarities between phases
8. Coherence Ratios
Intra/Inter ratios & PSI
Steps 1-5: Data Processing & Phase Detection
Raw Data:
dog (2s), cat (4s), lion (12s), tiger (12.5s), mouse (20s), rat (30s)
Word Vectors (300-dim):
dog: [0.12, -0.45, 0.78, ..., 0.23]
cat: [0.15, -0.42, 0.81, ..., 0.25]
Consecutive Similarities:
dog→cat: 0.82
cat→lion: 0.38
lion→tiger: 0.91
tiger→mouse: 0.31
mouse→rat: 0.85
Detected Phases:
Exploit: [dog, cat]
Explore: [cat→lion]
Exploit: [lion, tiger]
Explore: [tiger→mouse]
Exploit: [mouse, rat]
Similarity Plot:
dog→cat: 0.82 (EXPLOIT)
cat→lion: 0.38 (EXPLORE)
lion→tiger: 0.91 (EXPLOIT)
tiger→mouse: 0.31 (EXPLORE)
mouse→rat: 0.85 (EXPLOIT)
Threshold: ——————— (0.6)
Step 6: Intra-Phase Similarity Computation
Exploitation Phase: [dog, cat]
Similarity Matrix:
dog cat
dog 1.00 0.82
cat 0.82 1.00
Upper triangle: 0.82
μ_exploit_intra = 0.82
Exploration Phase: [cat→lion]
Transition similarity:
cat→lion = 0.38
μ_explore_intra = 0.38
Aggregation Across All Phases
For all exploitation phases:
μ_exploit_intra = mean(all pairwise similarities within exploit phases)
For all exploration phases:
μ_explore_intra = mean(all pairwise similarities within explore phases)
Step 7: Inter-Phase Similarity Computation
Step 7a: Phase Centroid Calculation
For each phase:
Centroid = mean(all word vectors)
Phase 1 (dog, cat):
c_1 = mean([dog_vec, cat_vec])
= [0.135, -0.435, 0.795, ...]
Phase 2 (lion, tiger):
c_2 = mean([lion_vec, tiger_vec])
= [0.180, -0.380, 0.820, ...]
Step 7b: Centroid Normalization
Normalize to unit length:
c_hat = c / ||c||
c_hat_1 = c_1 / ||c_1||
c_hat_2 = c_2 / ||c_2||
(For cosine similarity calculation)
Step 7c: Inter-Phase Similarity
Compute similarity between all phase centroid pairs:
S_inter(1,2) = c_hat_1 · c_hat_2 = 0.65
S_inter(1,3) = c_hat_1 · c_hat_3 = 0.58
S_inter(2,3) = c_hat_2 · c_hat_3 = 0.72
mu_inter = mean(all S_inter) = 0.65
Step 8: Coherence Ratios & Final Metrics
Coherence Ratios
Exploitation Coherence Ratio:
= μ_exploit_intra / μ_inter
= 0.82 / 0.65 = 1.26
Exploration Coherence Ratio:
= μ_explore_intra / μ_inter
= 0.38 / 0.65 = 0.58
Phase Separation Index (PSI)
PSI = (μ_exploit_intra + μ_explore_intra)/2 - μ_inter
= (0.82 + 0.38) / 2 - 0.65
Additional Metrics
• Exploitation %: 60%
• Exploration %: 40%
• Number of switches: 4
• Novelty score: 1.0
• Clustering coefficient: 0.75
Interpretation: Exploitation coherence ratio (1.26) > 1.0 indicates strong semantic clustering within exploitation phases. Exploration coherence ratio (0.58) < 1.0 indicates less clustering within exploration phases, consistent with switching between semantic domains.